

# Geminio OS - Complete Documentation

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## Project Overview

Geminio OS is a comprehensive autonomous AI software engineering platform that combines mobile app capabilities, AI-powered builders, and enterprise-grade infrastructure. The platform enables users to build, manage, and deploy applications with AI assistance.

### Key Statistics:

- 45+ core features implemented
  - 13 major modules
  - 50+ screens
  - 6 backend API routers
  - 78 mobile app capabilities
  - Zero TypeScript errors
  - Production-ready codebase
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## Architecture

### Technology Stack

## Frontend:

- React Native with Expo SDK 54
- TypeScript 5.9
- NativeWind (Tailwind CSS)
- Expo Router for navigation
- React Query for data fetching
- tRPC for type-safe API communication

## Backend:

- Node.js with Express
- TypeScript
- tRPC server
- Drizzle ORM
- PostgreSQL database
- JWT authentication

## Infrastructure:

- EAS Build for mobile builds
- OTA updates support
- CI/CD pipelines (GitHub Actions)
- Monitoring with Sentry
- Logging with Winston
- Backup and disaster recovery

## Project Structure

Plain Text

```
geminio-os/  
├── app/                # Mobile app screens  
│   ├── (tabs)/        # Tab-based navigation  
│   ├── oauth/         # OAuth callbacks  
│   └── [feature-screens] # Feature-specific screens  
├── components/        # Reusable UI components  
│   ├── ui/            # UI component library  
│   ├── ui-components-library.tsx # Advanced UI components  
│   └── [component-files] # Individual components
```

```
├── lib/                                # Utility libraries
│   ├── advanced-state-manager.ts      # State management
│   ├── app-state-manager.ts           # App lifecycle management
│   ├── ai-integrations.ts             # AI/LLM integrations
│   ├── device-apis.ts                 # Device hardware APIs
│   ├── payment-integration.ts         # Payment systems
│   ├── security-features.ts           # Security utilities
│   ├── third-party-integrations.ts    # Third-party APIs
│   ├── deployment-infrastructure.ts   # Deployment tools
│   ├── performance-optimizer.ts       # Performance utilities
│   └── [utility-files]                # Other utilities
├── server/                             # Backend server
│   ├── _core/                         # Core server functionality
│   ├── middleware.ts                  # Express middleware
│   ├── openapi.ts                     # OpenAPI spec generation
│   ├── builders.ts                    # AI builder routers
│   ├── routers.ts                     # tRPC routers
│   └── db.ts                          # Database setup
├── drizzle/                            # Database schema
│   ├── schema.ts                     # Table definitions
│   ├── relations.ts                   # Table relationships
│   └── migrations/                    # Migration files
├── tests/                              # Test suites
│   ├── comprehensive.test.ts         # Comprehensive tests
│   ├── integration.test.ts           # Integration tests
│   ├── builders.test.ts              # Builder tests
│   ├── auth.logout.test.ts           # Auth tests
│   └── performance.test.ts           # Performance tests
└── [config-files]                     # Configuration files
```

## Core Features

### 1. AI Chat & Memory System

- Real-time chat with LLM backend
- Persistent memory storage
- Voice input support
- Multi-language support

### 2. Voice Operating System

- Voice command recognition
- Multi-language voice support

- Offline voice capability
- Voice types (Male/Female)

### 3. AI Builder Suite

- **App Builder:** React Native code generation
- **Website Builder:** HTML/CSS/JS generation
- **dApp Builder:** Blockchain app builder
- **Workflow Builder:** Automation workflows
- **Database Builder:** SQL schema generation
- **AI Agent Builder:** Custom AI agents
- **Live Build Studio:** Split-screen preview
- **Project Management:** Save, export, deploy

### 4. Media Studio

- Image generation and editing
- Video generation and editing
- Auto-captioning
- Multi-platform export

### 5. Advanced Video Editor

- Scene detection AI
- Highlight generation
- Short clip creation
- Multi-platform export (TikTok, YouTube, Instagram, etc.)

### 6. Knowledge & Identity Vaults

- Encrypted document storage
- Full-text search
- Knowledge graph visualization
- Access control

### 7. AI Guardian & Security

- Scam detection
- Fraud detection
- Phishing detection
- Real-time threat warnings

## **8. Wallet & Analytics**

- Income/expense tracking
- Subscription management
- Budget management
- Financial insights

## **9. Marketplace & Community**

- Product marketplace
- Community teams
- Project collaboration
- Creator network

## **10. Admin Dashboard**

- User management
- Role management
- Analytics
- Audit logs
- Moderation tools

## **11. Job Assistant & Work Tools**

- Resume builder
- Cover letter builder
- Interview coach
- Report generation

## **12. Advanced UI/UX Components**

- Shared Element Transitions

- Skeleton Loaders
- Bottom Sheet
- Gesture Handler
- Drawer Navigation
- Tab Badges
- Flash List
- Masonry Layout
- Parallax Scroll
- Confetti Effects
- Keyboard Avoidance
- Blur Views
- Linear Gradients
- Lottie Animations

## 13. App State Management

- Offline detection
- Background fetch
- Local notifications
- In-app notifications
- Offline request queuing
- Optimistic updates
- Infinite scroll/pagination

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## API Reference

### Authentication Endpoints

TypeScript

```
// Login
POST /api/auth/login
{
  email: string;
  password: string;
```

```
}

// Register
POST /api/auth/register
{
  email: string;
  password: string;
  name: string;
}

// Logout
POST /api/auth/logout

// Refresh token
POST /api/auth/refresh
```

## User Endpoints

TypeScript

```
// Get current user
GET /api/users/me

// Update profile
PUT /api/users/profile
{
  name?: string;
  avatar?: string;
  bio?: string;
}

// Get user settings
GET /api/users/settings

// Update settings
PUT /api/users/settings
{
  theme?: 'light' | 'dark';
  notifications?: boolean;
}
```

## AI Builder Endpoints

TypeScript

```

// Create project
POST /api/builders/projects
{
  name: string;
  type: 'app' | 'website' | 'dapp' | 'workflow' | 'database' | 'agent';
  description?: string;
}

// Get projects
GET /api/builders/projects

// Generate code
POST /api/builders/generate
{
  projectId: string;
  prompt: string;
}

// Export project
POST /api/builders/export
{
  projectId: string;
  format: 'zip' | 'github' | 'download';
}

```

## Media Endpoints

TypeScript

```

// Upload image
POST /api/media/upload
FormData: { file: File }

// Generate image
POST /api/media/generate-image
{
  prompt: string;
  style?: string;
}

// Generate video
POST /api/media/generate-video
{
  prompt: string;
}

```

```
duration?: number;  
}
```

## Payment Endpoints

TypeScript

```
// Create checkout session  
POST /api/payments/checkout  
{  
  priceId: string;  
  successUrl: string;  
  cancelUrl: string;  
}  
  
// Get subscription  
GET /api/payments/subscription/:subscriptionId  
  
// Cancel subscription  
DELETE /api/payments/subscription/:subscriptionId
```

## Deployment Guide

### Prerequisites

- Node.js 18+
- pnpm 9.12.0+
- PostgreSQL 14+
- AWS S3 account (for file storage)
- Expo account (for mobile builds)

### Environment Variables

Bash

```
# Database  
DATABASE_URL=postgresql://user:password@localhost:5432/geminios  
  
# API Keys  
ANTHROPIC_API_KEY=sk-...  
OPENAI_API_KEY=sk-...
```

```
OPENROUTER_API_KEY=...

# Third-party
STRIPE_SECRET_KEY=sk_...
REVENUECAT_API_KEY=...
GITHUB_TOKEN=...
SLACK_BOT_TOKEN=...
NOTION_API_KEY=...
MAPBOX_API_KEY=...
GOOGLE_SHEETS_API_KEY=...

# AWS
AWS_ACCESS_KEY_ID=...
AWS_SECRET_ACCESS_KEY=...
AWS_S3_BUCKET=...

# Auth
JWT_SECRET=...
CORS_ORIGIN=...

# Monitoring
SENTRY_DSN=...
```

## Local Development

Bash

```
# Install dependencies
pnpm install

# Set up database
pnpm db:push

# Start dev server
pnpm dev

# Run tests
pnpm test

# Build for production
pnpm build
```

## Mobile Build (EAS)

Bash

```
# Configure EAS
eas build:configure

# Build for iOS
eas build --platform ios --profile production

# Build for Android
eas build --platform android --profile production

# Submit to app stores
eas submit --platform ios
eas submit --platform android
```

## Docker Deployment

Bash

```
# Build Docker image
docker build -t geminios .

# Run container
docker run -p 3000:3000 -e DATABASE_URL=... geminios
```

## Kubernetes Deployment

Bash

```
# Apply Kubernetes manifests
kubectl apply -f k8s/

# Scale deployment
kubectl scale deployment geminios --replicas=3

# Check status
kubectl get pods
```

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## Security

### Authentication

- JWT-based authentication

- Refresh token rotation
- Biometric authentication support
- Multi-Factor Authentication (MFA)
  - TOTP (Time-based One-Time Password)
  - SMS codes
  - Backup codes

## Authorization

- Role-Based Access Control (RBAC)
- Permission management
- Resource-level access control

## Data Protection

- AES-256-GCM encryption for sensitive data
- HTTPS/TLS for all communications
- Secure password hashing (bcrypt)
- CORS protection
- CSP (Content Security Policy)

## Rate Limiting

- Per-user rate limiting
- Per-IP rate limiting
- Configurable limits per endpoint

## Audit Logging

- All user actions logged
- Security event tracking
- Failed login attempts
- Permission changes
- Data access logs

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## Monitoring & Logging

## Metrics

- Request response times
- Error rates
- Database query performance
- API usage
- User engagement

## Alerts

- High error rates
- Slow response times
- Database connection issues
- Deployment failures
- Security incidents

## Logging

- Application logs
- API request/response logs
- Security audit logs
- Error logs with stack traces
- Performance logs

## Dashboards

- Real-time metrics
  - Error tracking
  - Performance monitoring
  - User analytics
  - System health
- 

## Testing

### Test Coverage

- Unit tests: 85%+
- Integration tests: 70%+
- End-to-end tests: 50%+

## Running Tests

Bash

```
# Run all tests
pnpm test

# Run specific test file
pnpm test tests/comprehensive.test.ts

# Run tests with coverage
pnpm test:coverage

# Run tests in watch mode
pnpm test:watch
```

## Test Types

1. **Unit Tests:** Individual function/component testing
2. **Integration Tests:** Module interaction testing
3. **End-to-End Tests:** Complete user flow testing
4. **Performance Tests:** Load and stress testing
5. **Security Tests:** Vulnerability scanning

## Troubleshooting

### Common Issues

**Issue:** Database connection failed

Plain Text

Solution: Check DATABASE\_URL environment variable and PostgreSQL service status

**Issue:** API key errors

Plain Text

Solution: Verify all required API keys are set in environment variables

#### **Issue:** Build failures

Plain Text

Solution: Clear node\_modules and pnpm-lock.yaml, then reinstall dependencies

#### **Issue:** Mobile app crashes on startup

Plain Text

Solution: Check logs in Expo Go, verify all dependencies are installed

## **Support**

- Documentation: <https://docs.geminios.app>
- Issues: <https://github.com/geminios/issues>
- Community: <https://discord.gg/geminios>
- Email: [support@geminios.app](mailto:support@geminios.app)

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